

Newton's Method

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1 Introduction

Newton's Method, also known as the Newton–Raphson Method, is one of the most efficient numerical techniques for solving nonlinear equations of the form

$$f(x) = 0.$$

Newton's Method starts from a single initial guess and uses the tangent line of the function to generate increasingly accurate approximations to the root.

The main idea is straightforward. Suppose the current approximation is x_n . Instead of estimating the root using two previous points as in the Secant Method, Newton's Method computes the tangent line at $(x_n, f(x_n))$. Since the tangent line is a good local approximation of the function, its intersection with the x-axis is expected to be a better estimate of the root. Repeating this process produces a sequence

$$x_0, x_1, x_2, \dots$$

that converges toward the solution.

2 Mathematical Formulation

2.1 Tangent Line

The fundamental idea of Newton's Method is based on the tangent line of a function. The derivative of a differentiable function at a point is defined as the slope of the tangent line at that point. For a differentiable function $y = f(x)$, the slope of the tangent line at the point $(a, f(a))$ is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

where the limit is assumed to exist. Since the tangent line passes through the point $(a, f(a))$ and has slope $f'(a)$, its equation is given by

$$y - f(a) = f'(a)(x - a).$$

2.2 Derivation of Newton's Iteration Formula

Suppose x_n is the current approximation to a root of the nonlinear equation $f(x) = 0$. At the point $(x_n, f(x_n))$, the equation of the tangent line is

$$y - f(x_n) = f'(x_n)(x - x_n).$$

Starting from the current approximation x_n , Newton's Method replaces the nonlinear function with its tangent line, which provides a linear approximation to the curve in a neighborhood of x_n . The next approximation, denoted by x_{n+1} , is chosen as the x-intercept of the tangent line because

it is expected to be a better estimate of the root. The x-intercept is obtained by setting $y = 0$. Substituting $y = 0$ into the tangent-line equation gives

$$0 - f(x_n) = f'(x_n)(x_{n+1} - x_n).$$

Rearranging the equation gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

This equation is known as the Newton iteration formula. Starting from an initial guess x_0 , the iteration formula is applied repeatedly to generate the sequence $x_0, x_1, x_2, \dots$, which converges to the root under suitable conditions.

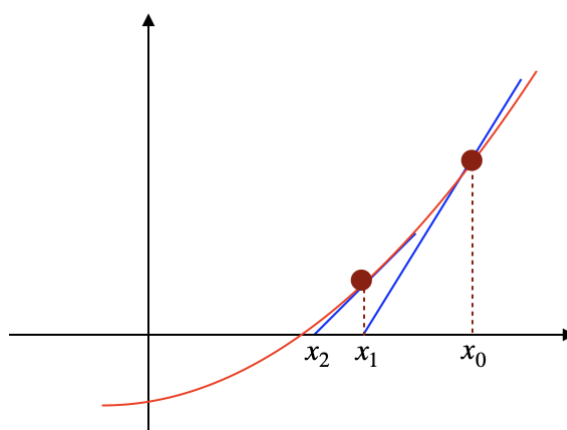


Figure 1: Newton's Method

3 Algorithm Flow

Starting from an initial approximation x_0 , Newton's Method repeatedly updates the current approximation using the Newton iteration formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$. At each iteration, both the function value $f(x_n)$ and its derivative $f'(x_n)$ are evaluated to compute the next approximation. After obtaining x_{n+1} , the function value $f(x_{n+1})$ is evaluated. If the difference is smaller than the tolerance, the iteration is terminated. Otherwise, the newly computed approximation becomes the starting point for the next iteration. Since Newton's Method requires the derivative at each iteration, the algorithm checks that $f'(x) \neq 0$ to avoid division by zero.

Algorithm 1 Newton's Method

Input: Function $f(x)$, Derivative $f'(x)$, Initial guess x_0 , Tolerance ϵ , Maximum iterations N

Output: Approximate root x

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for  $K = 1$  to  $N$  do
    if  $|f'(x_0)| < \delta$  then
        return failure
    end if
     $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ 
    if  $|f(x_1)| < \epsilon$  then
        return  $x_1$ 
    end if
     $x_0 = x_1$ 
end for
return  $x_1$ 

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4 Convergence Analysis

4.1 Taylor's Theorem with Lagrange Remainder

Taylor's theorem approximates a smooth function by a polynomial constructed from its derivatives at a given point. Since a finite polynomial cannot exactly represent most functions, a remainder term is introduced to measure the approximation error.

Suppose f is $(n+1)$ -times continuously differentiable on an interval containing a and x . Then Taylor's theorem states that

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^n(a)}{n!}(x - a)^n + R_n(x),$$

where $R_n(x)$ is called **remainder term**.

One common expression for the remainder is the **Lagrange remainder**:

$$R_n(x) = \frac{f^{n+1}(\xi)}{(n+1)!}(x - a)^{n+1},$$

where ξ is some point lying between a and x .

4.2 Quadratic Convergence

One of the major advantages of Newton's Method is its fast convergence near a simple root. Under suitable conditions, Newton's Method has quadratic convergence.

Assume that f is twice continuously differentiable in a neighborhood of the simple root x^* , where $f(x^*) = 0$ and $f'(x^*) \neq 0$. Let the error at iteration n be $e_n = x_n - x^*$. To determine

how the error changes between iterations, apply Taylor's theorem to $f(x^*)$ about the current approximation x_n :

$$f(x^*) = f(x_n) + f'(x_n)(x^* - x_n) + \frac{f''(\xi_n)}{2}(x^* - x_n)^2,$$

where ξ_n lies between x_n and x^* . Since x^* is a root, $f(x^*) = 0$. Therefore,

$$0 = f(x_n) + f'(x_n)(x^* - x_n) + \frac{f''(\xi_n)}{2}(x^* - x_n)^2.$$

Rearranging,

$$f(x_n) = f'(x_n)(x_n - x^*) - \frac{f''(\xi_n)}{2}(x_n - x^*)^2.$$

Dividing both sides by $f'(x_n)$,

$$\frac{f(x_n)}{f'(x_n)} = (x_n - x^*) - \frac{f''(\xi_n)}{2f'(x_n)}(x_n - x^*)^2.$$

Recall Newton's iteration formula,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Substituting the previous expression gives

$$x_{n+1} = x_n - [(x_n - x^*) - \frac{f''(\xi_n)}{2f'(x_n)}(x_n - x^*)^2].$$

Hence,

$$x_{n+1} = x^* + \frac{f''(\xi_n)}{2f'(x_n)}(x_n - x^*)^2.$$

Subtracting x^* from both sides,

$$x_{n+1} - x^* = \frac{f''(\xi_n)}{2f'(x_n)}(x_n - x^*)^2.$$

Using the error notation,

$$e_{n+1} = \frac{f''(\xi_n)}{2f'(x_n)}e_n^2$$

As $n \rightarrow \infty$, $x_n \rightarrow x^*$. Consequently, $\xi_n \rightarrow x^*$ by continuity of f' and f'' , $\frac{f''(\xi_n)}{2f'(x_n)} \rightarrow \frac{f''(x^*)}{2f'(x^*)}$. Therefore,

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^2} = \left| \frac{f''(x^*)}{2f'(x^*)} \right|,$$

provided this constant is finite and nonzero.

Comparing this with the general definition of convergence order,

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^p} = C, \quad 0 < C < \infty$$

we obtain $p = 2$.

Therefore, once the iterates are sufficiently close to the root, the error at each iteration is approximately proportional to the **square** of the previous error. Consequently, the number of correct decimal digits approximately doubles with each iteration, making Newton's Method significantly faster than linearly convergent methods.

5 Numerical Example

To illustrate the Newton's Method, consider the nonlinear equation

$$f(x) = x^2 - 2.$$

The exact root is

$$x = \sqrt{2} \approx 1.41421356..$$

Choose the initial guess

$$x_0 = 1.0.$$

Applying the Newton's Method produces the following sequence of approximations.

Table 1: $f(x) = x^2 - 2$		
Iteration	Approximation	$f(x)$
1	1.5	2.5×10^{-1}
2	1.41667	6.94×10^{-3}
3	1.41422	6.0073×10^{-6}
4	1.41421	4.51084×10^{-12}

After 4 iterations, the approximate root is 1.41421.

The convergence behavior is clearly observed: $10^{-3} \rightarrow 10^{-6} \rightarrow 10^{-12}$.